2023 SSMO Team Round

SMO Team

TR 1: Let (a, b, c, d) be a permutation of $(2, 0, 2, 3)$. Find the largest possible value of $a^b + b^c + c^d + d^a$.

TR 2: A plane and a car start moving at the same time. The car moves northbound at 60 miles per hour. The plane moves northeast and increases in altitude at an angle of 30°. Let s be the speed in feet per second that the plane must fly at to move north at the same speed as the car. Find $3s^2$.

TR 3: Let ABC be a triangle such that $AB = 4\sqrt{2}$, $BC = 5\sqrt{2}$, and $AC = \sqrt{82}$. Let ω be the circumcircle of $\triangle ABC$. Let D be on the circle such that $\overline{BD} \perp \overline{AC}$. Let E be the point diametrically opposite B. Let F be the point diametrically opposite D. The area of the quadrilateral $ADEF$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

TR 4: Find the sum of all primes p such that $p \mid (2023^{p^2} + (p-1)! + 2^{p^4})$.

TR 5: Joshy is playing a game with a dartboard that has two sections. If Joshy hits the first section, he gets 20 points, and if he hits the second section, he gets 23 points. Assume Joshy always hits one of the two sections. Let a be the largest score Joshy cannot achieve. Let b be the number of positive integer scores Joshy cannot achieve. Let c be the number of ordered pairs (x, y) of nonnegative integers with $20x + 23y = 2023$. Find $(a - b)c$.

TR 6: Suppose that a, b, and c are positive reals satisfying

$$(a^3 + 4)(b^3 + 6)(c^3 + 8) = 8(a + b + c)^3.$$

Find the sum of all possible values of $\frac{bc}{a^2}$. (If there are none, give an answer of 0.)

TR 7: Let $S = \{1, 2, 3, 4, \ldots, 23\}$ and let sets $A, B, C \subseteq S$ be chosen independently and uniformly at random. The probability that $|A| + |B| = |C|$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Let 2^a be the largest power of 2 such that $2^a \mid n$. Find a.

TR 8: Three rabbits run away from the origin at the same speed and constant velocity such that the angle between any two rabbits' directions is 120°. After 12 seconds, a hunter with a speed $\sqrt{7}$ times that of the rabbits runs from the origin. The minimal time in seconds, measured from when the hunter starts running, needed for her to catch all three rabbits can be expressed as $a + b\sqrt{c}$, where a, b, and c are positive integers and c is not divisible by the square of any prime. Find $a + b + c$.

TR 9: Let B, K, and R be the total number of possible moves for a bishop, knight, or rook, respectively, summed over all positions of a 9 by 9 grid. Find $B+K+R$. (A bishop moves along diagonals, a rook moves along rows and columns, and a knight moves in the form of an L: two squares in one direction and one square in a perpendicular direction.)

TR 10: There is an infinite row of cars. Each car independently has a $\frac{1}{3}$ chance of being high quality and a $\frac{2}{3}$ chance of being low quality. John goes down the row of cars buying high-quality cars. However, after John sees 3 low-quality cars, he gives up on buying additional cars. The probability that he buys at least 5 cars before giving up can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

TR 11: Let $ABCD$ be a cyclic quadrilateral such that AC is a diameter. Let P be the orthocenter of ABD. Let X be the intersection of lines AB and CD, and let Y be the intersection of lines AD and BC. If $AB=8$, $BC=1$, and $CD=4$, then $\frac{[CBPD]}{[AXY]}$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

TR 12: Let T be the set of rationals of the form $\frac{a}{2^b}$ for nonnegative a and b. Define the function $f\colon T\to\mathbb{Z}$ such that, for $t=\frac{a}{2^b}$ such that b is minimal, we have that

$$f\left(\frac{a}{2^b}\right)=\begin{cases}1 & b=0\\ f\left(\frac{a-1}{2^b}\right)+f\left(\frac{a+1}{2^b}\right) & b\neq 0\end{cases}$$

The value of

$$\sum_{i=0}^{2^{10}-1}\frac{f\left(\frac{i+1}{2^{10}}\right)}{f\left(\frac{i}{2^{10}}\right)}$$

can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

TR 13: Let $D(n)$ denote the product of all divisors of n. Let $P(i,j)$ denote the set of all positive integers that are both a multiple of i and a factor of j. Let

$$F(x)=-\sqrt{\left|\log_{10}\left(\frac{D(10^x)}{\prod_{\omega\in P(10,10^{x+1})}\omega}\right)\right|}\quad\text{and}\quad G(n)=\sqrt[n-1]{\prod_{i=2}^{n}10^{F(i)}}.$$

The value of $\sum_{k=2}^{\infty}G(k)$ can be expressed as $\frac{a+b\sqrt{c}}{d}$, where a, b, c, and d are positive integers, c is not divisible by the square of any prime, and $\gcd(a,b,d)=1$. Find the value of $a+b+c+d$.

TR 14: For how many primes $p<100$ does there exist a positive integer k such that p divides 10^k+1?

TR 15: Consider a piece of paper in the shape of a regular pentagon with side length 2. We fold it in half. We then fold it again so that the two endpoints of its longest side coincide. The area of the folded figure can be expressed as $\frac{1}{a}\sqrt{b+c\sqrt{5}}$, where a, b, c are positive integers with a minimized. Find $a+b+c$.

